

MATH310: Multiple Comparisons Project

Jo's Weighted Anchor Method

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1. Method Correction

After revision, the intuition behind Jo's Weighted Anchor method is that we first calculate an "anchor" – the corrected alpha level such that even if we were to perform all pair-wise t-tests, we would still retain our desired FWER. This is actually the concept of the Sidak Anchor.

Then our "budget" for allocating our significance levels are weighted based on how "uncertain" a p-value is. For p-values that are close to 0 or 1, we already know what the results will be. Hence, we take alpha budget from those p-values and re-distribute to values that are more uncertain in what the results will be, namely, the p-values that are close to the significance level.

Because we have a varying significance level for each pair-wise comparison, the method mimics Holm-Bonferroni to some degree.

Revised Algorithm:

- (a) First, we will store information for each pair of means. Compute the difference in means, conduct a two-sample t-test for each pair, and store the p-value.
- (b) Calculate anchor α_1 with the following formula:

$$\alpha_1 = 1 - (1 - \alpha_0)^{\frac{1}{m}}$$

Where α_0 = desired FWER and m = number of pairwise comparisons.

- (c) Calculate the uncertainty of each p-value, p_i which is given by:

$$\text{uncertainty} = \frac{1}{\text{distance}} = \frac{1}{|\log \frac{\alpha_1}{p_i}|}$$

We define distance in an unconventional way: Essentially, we are first calculating what proportional distance our p-value is to our anchor, then taking the logarithm to "tame" exponential differences, and taking the absolute value so that distances are the same in either direction.

To calculate uncertainty, we take the reciprocal so that the closer a p-value is to the anchor, the more uncertain we are.

- (d) Then we calculate the weight of each p-value, which is just the proportion of uncertainty a p-value contributes to the total uncertainty.
- (e) For p-values $> \alpha_1$ (The p-values that don't get rejected), add budget to the anchor that is proportional to its weight but scaled by a factor η , as shown below:

$$\text{new threshold} = \alpha_1 + \text{weight} * \eta$$

- (f) For p-values $\leq \alpha_1$ (The p-values that get rejected), subtract budget from the anchor that is proportional to its weight but scaled by a factor η , as shown below:

$$\text{new threshold} = \alpha_1 - \min\left(\frac{1}{\text{weight}}, \left(\frac{\alpha_1}{\eta} - \epsilon\right) * \eta\right)$$

*See the discussion below for more explanation on this equation.

(g) Based on these newly allocated thresholds, determine significance of each pairwise comparison.

Justification:

The reason why this method is justified is that first, we begin with the sidak anchor, which by definition, will stay within the desired FWER. It is true that the way we take and distribute budget from different p-values deviates from this mathematical guarantee, but this is why we use the empirical calibration method for eta, which ensures that we use the correct hyperparameter to maintain the desired FWER. One might suspect an issue of overfitting the data, but the eta is specific to the size I and J , so the same eta is generalizable to other datasets of the same size I and J .

To give more context on step (f), we are trying to ensure that our new threshold is always greater than 0. Hence, although the value we really want to subtract from α_1 is $\frac{1}{\text{weight}}$, we must incorporate the $(\frac{\alpha_1}{\eta} - \epsilon) * \eta$ factor (for some $\epsilon > 0$), because

$$\alpha_1 - (\frac{\alpha_1}{\eta} - \epsilon) * \eta = \alpha_1 - \alpha_1 + \epsilon\eta = \epsilon\eta > 0$$

Because we always control parameters so that $\eta > 0$. Therefore, taking the minimum of the two values will ensure that our new threshold is always positive.

Further, we are using $\frac{1}{\text{weight}}$ and not *weight* as we did in step (e), because we want to de-allocate more budget from more certain values, which in this case, are the smaller weighted values. This was the other way around for step (e).

As an analogy, step (e) is distributing resources to more urgent populations and step (f) is taxing the rich.

Key Revisions

Bugs Fixed

- (a) We now correct the constraint equation to use the product of (1- alpha) instead of the product of alphas as before.
- (b) We use the correct degrees of freedom in our method.
- (c) Now we report Standard Error (SE) with the FWER simulations.
- (d) With the new method, we don't have the Scenario B outlier with FWER = 0.0764

Implemented Suggestions

- (a) We reconsidered the weighting direction so that we give more budget to uncertain p-values rather than to the comparisons with large observed differences.
- (b) We implement an empirical calibration method for eta, much like the hyperparameter tuning in a machine learning algorithm.
- (c) In order to keep the method simple, we save the "step-down" structure and incorporating correlation structure for a future reiteration of the method.

2. Implement Your Method in R

Refer to "mc_project budget (main)" .r for function implementation and comments. Below is the demonstration using Example 11.5 Oil Filters Data with pre-calibrated $\eta = 0.001$:

	group1	group2	diff	pval	dist	uncertainty	weights	threshold	significant
13	1	4	0.2	0.1604268	1.165815	0.8577693	0.3218561	0.0054381	FALSE
19	3	5	0.2	0.1604268	1.165815	0.8577693	0.3218561	0.0054381	FALSE
15	2	3	0.5	0.0009317	3.982781	0.2510809	0.0942117	0.0039504	TRUE
16	2	4	0.5	0.0009317	3.982781	0.2510809	0.0942117	0.0039504	TRUE
11	1	2	0.7	0.0000116	8.366388	0.1195259	0.0448491	0.0039504	TRUE
17	2	5	0.7	0.0000116	8.366388	0.1195259	0.0448491	0.0039504	TRUE
18	3	4	1.0	0.0000000	15.279232	0.0654483	0.0245578	0.0039504	TRUE
12	1	3	1.2	0.0000000	19.758858	0.0506102	0.0189902	0.0039504	TRUE
10	4	5	1.2	0.0000000	19.758858	0.0506102	0.0189902	0.0039504	TRUE
14	1	5	1.4	0.0000000	24.009794	0.0416497	0.0156280	0.0039504	TRUE

3. Validate via FWER Simulation (Re-run Simulation)

Below is the estimated FWER for each scenario.

Scenario	I	J	FWER	SE
IA	3	10	0.0484	0.0021461
IB	5	10	0.0569	0.0023165
IC	7	10	0.0468	0.0021121
ID	5	5	0.0535	0.0022503
IE	5	20	0.0500	0.0021794

All scenarios have FWER in between 0.045 and 0.06, which is well within the target range of 0.03 and 0.07.

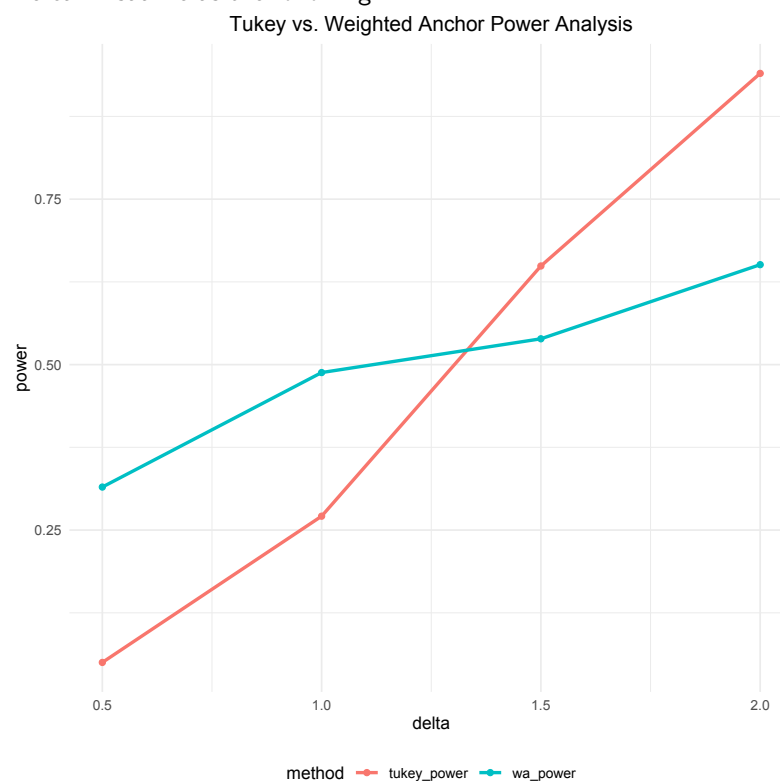
Refer to "mc_project budget (main)".r for corrected code.

4. Power Analysis (Re-run Simulation)

Below is the power for the Weighted Anchor method versus Tukey's HSD for various effect sizes.

delta	wa_power	tukey_power
0.5	0.315	0.050
1.0	0.488	0.271
1.5	0.539	0.649
2.0	0.651	0.940

We can visualize as the following:



We see that the Weighted Anchors method is more powerful than Tukey's HSD for effect sizes 0.5 and 1.0, but not for 1.5 and 2.0. Hence, the Weighted Anchors method is better suited to detect real, but smaller differences in means as compared to Tukey's HSD.

Refer to "mc_project budget (main)".r for corrected code.